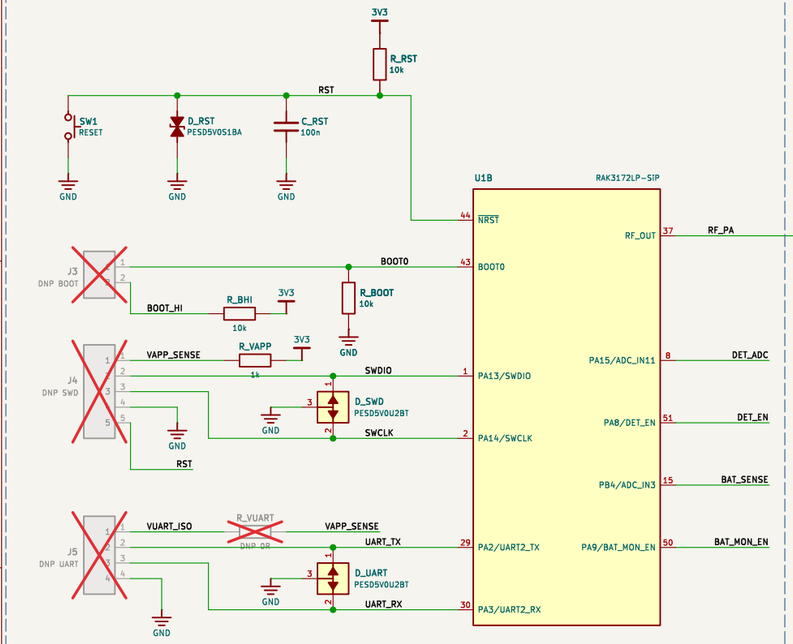
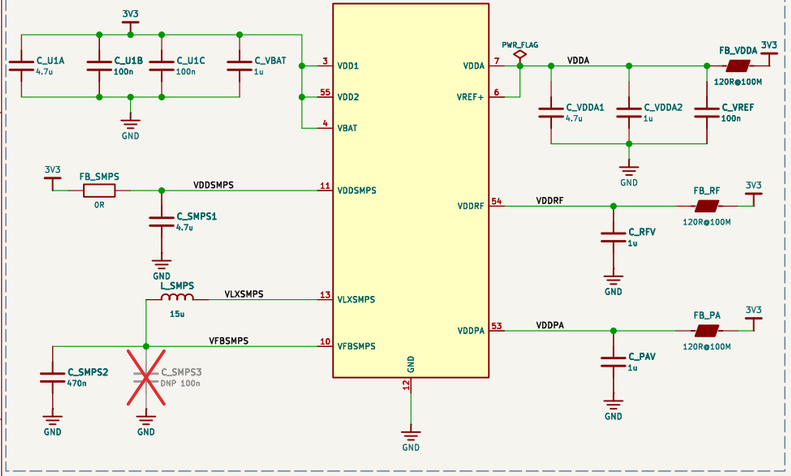
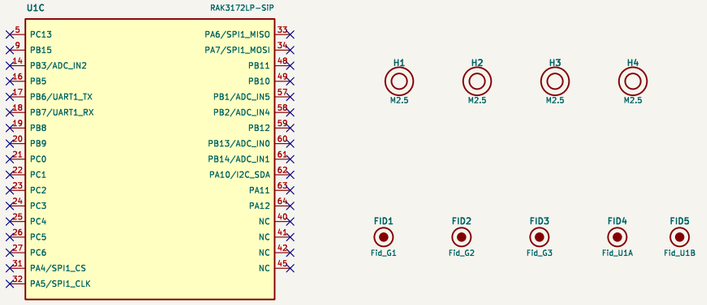
1. MCU CONTROL, PROTECTED DEBUG & RF LAUNCH (U1B)



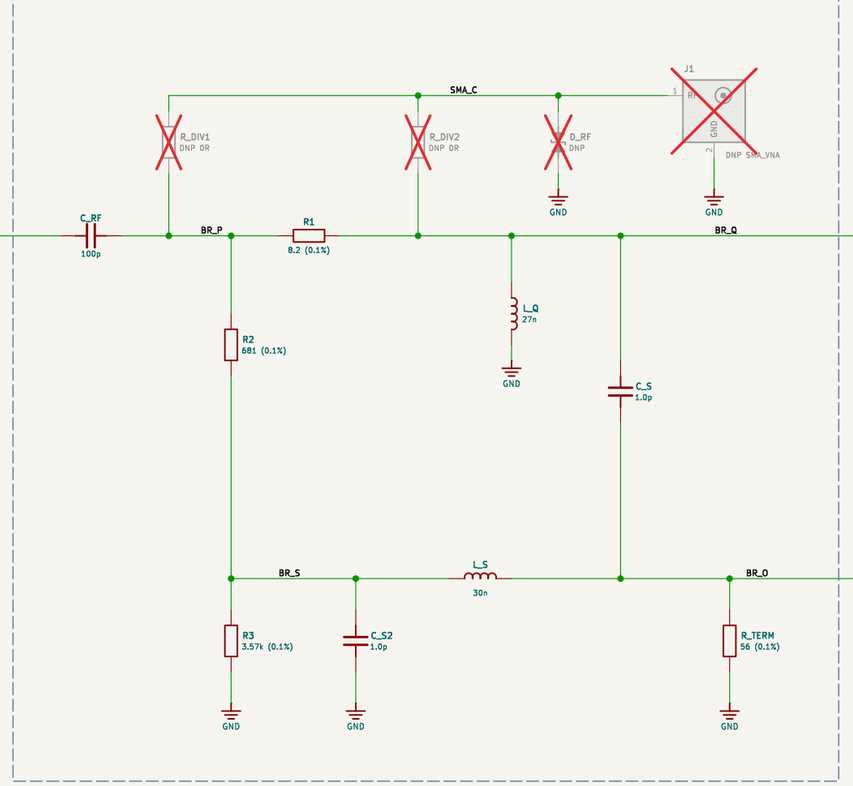
2. SIP POWER, ISOLATED VDDA & 15 uH BUCK SMPS (U1A)



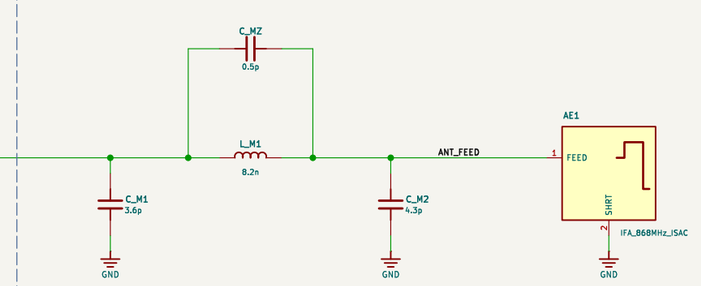
3. UNUSED LGA-73 I/O BANK (U1C) & MECHANICALS



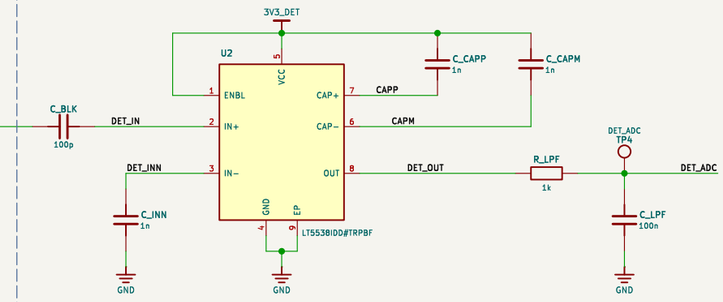
4. 868 MHz RETURN-LOSS BRIDGE & VNA CALIBRATION PORT



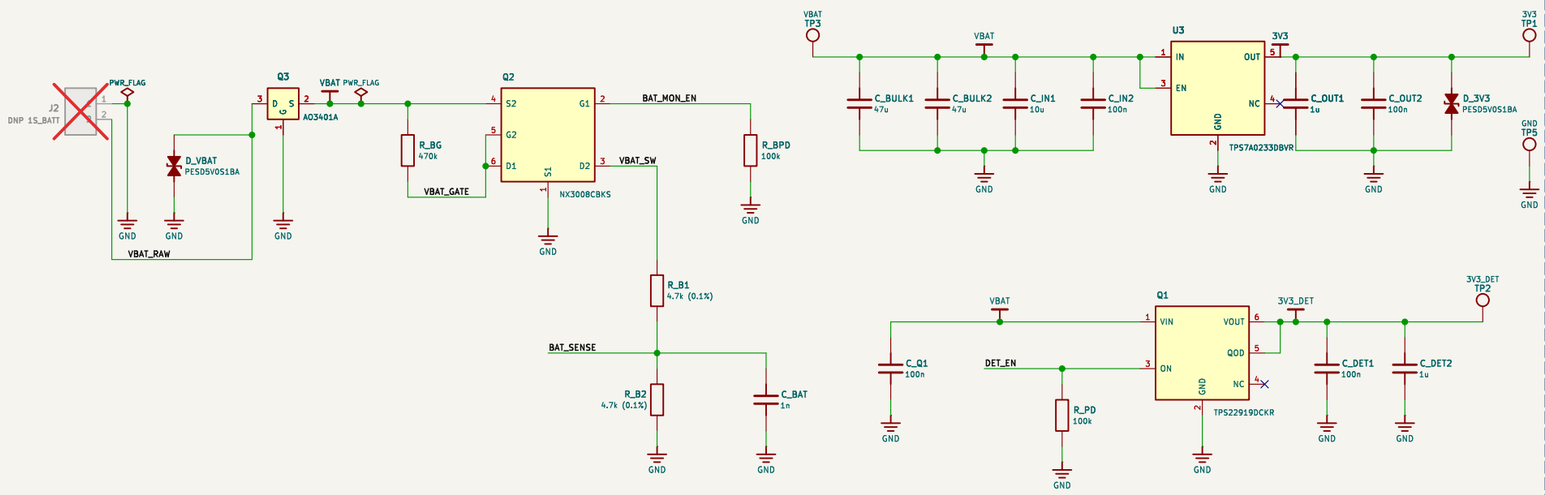
5. ELLIPTIC HARMONIC PI-FILTER & FOLDED IFA-SENSOR



6. LT5538 75 dB RF LOG DETECTOR & VIDEO LPF



7. 1S BATTERY INPUT, GATED VBAT SENSE, 3.3 V LDO & DETECTOR RAIL



HARDWARE DESIGN & MANUFACTURING NOTES (REV E.5):
1. POWER: 1S LiPo / Li-SOCl2 input (J2: 1=GND, 2=+VBAT, 3.0–4.35 V DC max). Q3 (AO3401A) RPP feeds U3 (TPS7A0233, 25 nA iq) and Q1 (TPS22919) -> U2.
2. ANALOG & SMPS: FB_VDDA isolates VDDA/VREF+; L_SMPS = 15 uH power inductor (Murata LQH2MCN150K02L, Isat >= 300 mA); internal 3.0 V TCXO enabled.
3. RF & FIELD SKU: 51.9 Ohm CPWG spine, elliptic harmonic pi-filter, bare-ENIG sensing comb; J1..J5 DNP on sealed field units (populate for bench/VNA).

Power: 1S LiPo/Li-SOCl2 (3.0–4.35V), AO3401A RPP, TPS7A0233 3V3
RF Sensor: 8.2R Return-Loss Bridge + ADI LT5538 Log Detector
Antenna: 866.5 MHz Folded IFA + Elliptic Pi (3.6p/8.2n||0.5p/4.3p)
MCU/Radio: RAK3172LP-SiP (STM32WLE5JC, 3.0V TCXO, 15uH SMPS)
RF & Embedded Systems Engineering

Sheet: /
File: ISAC_868_KiCad9.kicad_sch

Title: Dual-Function 868 MHz PCB Antenna-Sensor (ISAC)

Size: A3	Date: 2026-11-06	**Rev: E.5**
KiCad E.D.A. 9.0.8		Id: 1/1